

LEDGER · 2026-08-27 · MICROALGAE BIOTECHNOLOGY

C-Phycocyanin via Heterotrophic Cultivation of *Galdieria sulphuraria*

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Spirulina Extract (C-Phycocyanin, E25 Grade)
Entry application	low-pH, pasteurized functional beverages and sports drinks
Measured advantage	Maintains stability down to pH 2.8 and up to 60°C, whereas Spirulina E25 denatures above 45°C and irreversibly precipitates below pH 4.0.
Time to first revenue	48 months
Regulatory risk	Moderate — qualification costs reported (\$850,000 estimated)

Why it failed

- Voided by: Capital and time to revenue, Economics undisclosed

Assessment

The concept fails venture criteria due to severely negative gross margins, excessive CapEx (\$1.25M), and extended regulatory delays (48 months to revenue). Heterotrophic carbon

consumption leads to an OpEx of \$405/kg against an incumbent market price of \$130/kg. Technical stability improvements cannot overcome these broken unit economics.

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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